Thermal-Headroom Routing for Efficient Robot Perception in Extreme Heat

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***Abstract*—A robot working in a Phoenix summer runs its detector on a small embedded processor that heats itself. Once that processor hits its thermal limit, the frequency governor throttles it and every detector configuration gets slower at the same time. Adaptive inference routers, which pick a model size and input resolution for each frame, usually treat this slowdown as outside interference. In hot weather the router causes it. We model the effect with a lumped RC thermal model wired in closed loop to a deadline-constrained router over four YOLO11 configurations. We also propose Thermal-Headroom Routing (THR), which forecasts the processor temperature in closed form before committing to a configuration and skips any configuration that would push the chip past its throttle threshold. We tested it on seven real traces with camera degradation (off-road driving, rain, snow, fog, night, and lens soiling), eight ambient temperatures, and five seeds. At 45 °C and 55 °C THR never throttled. A router driven only by latency belief spent 11 to 38% of each mission throttled. THR also lowered the deadline-miss rate by 3.9 to 7.4 percentage points and energy per on-time frame by 11 to 13%, and it costs about 7 μs per frame. At 35 °C and below it makes the same decisions as the baseline. A control that gets the same share of heavy-model frames, but no temperature reading, reaches the same miss rate. The miss-rate gain therefore comes from the amount of heavy inference the router allows. What the thermal forecast adds is a way to find that amount online, at any ambient temperature, while keeping throttling at zero. Code and outputs are public.[1]**

***Index Terms*—efficient inference, adaptive model selection, thermal throttling, edge AI, robot perception, extreme environments**

1. Introduction

Consider a Jetson-class module in a sealed enclosure on a robot in the Sonoran Desert. The air is at 45 °C. If the perception stack runs its largest detector on every frame, the chip reaches its throttle limit in about a minute. The governor then down-clocks it, and the small models slow down along with the large one until the chip cools [1]. In our simulations a policy that always runs the large model goes blind for 12.5 s at a stretch at 45 °C, and for 91 s at 55 °C (we count a blackout as a run of consecutive deadline misses).

Adaptive inference is the usual answer to a tight latency budget. Early-exit networks and routers choose cheaper computation per input or per frame [2]–[4], and some also watch current compute load. All of them react to latency after it has risen. In heat that reaction comes late, because the slowdown was set up by earlier choices and lasts for tens of seconds once it starts. Thermal-aware scheduling for multiprocessors has dealt with this kind of delayed feedback for years [5], [6], and planetary rovers plan compute around thermal and power limits [7]. Per-frame model selection has not used those ideas. Belief-space perception routing [8], which this work extends, tracks camera-fault and compute-contention beliefs together but leaves thermal effects out of its contention model.

Our contributions: (i) a closed-loop model of thermal contention for adaptive inference, in which the router's configuration choice sets the power input to an RC thermal node with a hysteretic throttle (§3); (ii) THR, a closed-form rule that keeps the forecast temperature under the throttle limit at about 7 μs per frame and works alongside fault-aware routing [8] (§4); (iii) an evaluation on seven real degraded-camera traces over eight ambient temperatures, with a cool-weather negative control and a matched-budget control in the spirit of the matched-input controls in [9], which tells apart the effect of how much heavy inference runs from the effect of when it runs (§6); and (iv) power, energy and accuracy accounting for every policy, plus a test of THR with a wrong thermal model.

2. Related Work

Adaptive and efficient inference. Early-exit networks [2], [3], dynamic transformers [4] and early exiting for driving [10]–[12] vary computation with the input. Modality gating [13]–[15] decides which sensors to process at all. These methods treat the compute budget as fixed or as something they observe. None of them models the budget as a state their own decisions change.

Sensor- and compute-aware routing. Belief-space perception routing [8] tracks blur, illumination and occlusion faults

1. https://github.com/ImSpxrsh/thermal-headroom-routing

per camera channel with switching HMMs, keeps a latency-based contention belief, and fuses the two with a noisy-OR term. When faults and contention were made to co-occur, that fusion cut deadline misses by 1.1 to 9.4 percentage points. Its contention signal comes from outside the router and has two states. We reuse its fault beliefs, frontier and traces and add a thermal plant that the router drives.

Thermal management. HotSpot-style RC models [6] and temperature-aware MPSoC scheduling [5] predict hot spots and schedule around them. Measurements on commercial phone SoCs show throttling dominating sustained performance [1]. We apply the same forecasting idea to per-frame model selection, where the thing being chosen is a point on the accuracy-latency curve.

Datasets. The traces come from TartanDrive off-road driving [16], [17], RADIATE adverse weather [18] and WoodScape fisheye soiling [19].

3. System Model

Frontier and deadline. There are four configurations, from most to least expensive: C1 is YOLO11x at 1280 px, C2 is YOLO11x at 640, C3 is YOLO11n at 1280 and C4 is YOLO11n at 640. Their per-frame latencies were profiled under normal and contended compute in [8]. The deadline D is C1's median latency under normal compute, between 211 and 341 ms depending on the trace. C1 therefore meets D about half the time on a cool chip and almost never on a slowed one. C2, C3 and C4 always meet it. Accuracy is the IoU-matched F1 agreement between a configuration's detections and C1's on the same frame. The earlier study found that this pseudo-ground truth ranks configurations differently from real annotations [8], so we treat every accuracy number here as an internal consistency check. Deadline-miss, throttling and energy results do not depend on it.

Thermal plant. We model the SoC as one thermal node with time constant τ and junction-to-ambient resistance R_{th}:

$$T_{t+1} = T_t + \tfrac{\Delta t}{\tau}\big(T_{\mathrm{amb}} + R_{\mathrm{th}}\,P_t - T_t\big). \qquad (1)$$

Power depends on the configuration c_t served at frame t: $P_t = s_t\,[P_{\mathrm{idle}} + (P_{\max} - P_{\mathrm{idle}})\,u(c_t)]$. Here $u(c) = \min(1, \tilde{\ell}_c/D)$ is the configuration's median normal-compute latency as a fraction of the deadline, and s_t is 0.6 while throttled and 1 otherwise. The governor starts throttling at $T \geq T_{\mathrm{hi}}$ and stops at $T \leq T_{\mathrm{lo}}$. While it throttles, every configuration's latency is drawn from its profiled contended distribution, whose p95 is about $2.5\times$ the normal one. Table 1 lists the parameters. They are typical of a Jetson Orin module and were chosen before any run, but we did not measure them. Running C1 nonstop settles at $T_{\mathrm{amb}} + 50\,^{\circ}\mathrm{C}$, so throttling starts once the air is above 40 °C. The repository includes a Jetson harness that logs `tegrastats` temperature and power, which can supply measured values.

Closed loop. In [8] contention followed a fixed schedule, so every policy could be scored against the same pre-drawn timeline. Here, whether the chip is throttled at frame t depends on what the router chose before t, so each policy has to run in closed loop with the plant. For a given seed, all policies see the same pre-drawn latency samples (normal and contended) and the same sensor noise.

TABLE 1. THERMAL-PLANT PARAMETERS (TYPICAL VALUES, FIXED BEFORE ANY RUN).

Δt (frame)	0.1 s	P_{idle} / $P_{\max}$	5 / 25 W
τ	30 s	T_{hi} / T_{lo}	90 / 85 °C
R_{th}	2.0 °C/W	throttled power	0.6×

4. Thermal-Headroom Routing

All policies use the objective from [8]. They predict a compute state, keep the configurations whose median latency in that state meets D, and serve the one with the highest expected accuracy given the camera-fault belief b_f. A 10-frame dwell keeps them from switching back and forth. They differ only in the inputs they use.

Online compute belief. In closed loop the router only learns the latency of the configuration it served, and only one frame later. We run a two-state forward filter (stay probabilities 0.95 and 0.90) on one binary observation: whether the last latency exceeded $1.5\times$ that configuration's normal median. The filter output is b_c.

THR. Let $\hat{T}$ be the on-die temperature reading (we add Gaussian noise with $\sigma = 1\,^{\circ}\mathrm{C}$). If the router held configuration c for H frames, Eq. (1) gives the temperature in closed form:

$$\hat{T}_H(c) = T_\infty(c) + \big(\hat{T} - T_\infty(c)\big)\big(1 - \tfrac{\Delta t}{\tau}\big)^H, \qquad (2)$$

with $T_\infty(c) = T_{\mathrm{amb}} + R_{\mathrm{th}}P(c)$. THR drops c from consideration when $\hat{T}_H(c) \geq T_{\mathrm{hi}} - m$, using $H = 50$ frames and a margin $m = 2\,^{\circ}\mathrm{C}$. C4 is always allowed, so some configuration is always available. Feasibility and accuracy selection then run as usual. Each frame costs four evaluations of Eq. (2). With the belief update and selection included, we measured 7.3 μs per frame in Python on one Apple M5 core, under 0.004% of one C1 inference.

Joint + THR. Camera faults sometimes come with extra compute load, for example when a degraded camera makes localization work harder. For that case THR can be combined with the noisy-OR coupling of [8], $p_{\mathrm{cont}} = 1 - (1 - b_c)(1 - \kappa b_f)$, where κ is fit on separate calibration draws.

Baselines. Static C1 and Static C4 always serve one configuration. Reactive looks at latency only and ignores the camera. Belief is the decoupled router of [8] using our online b_c and no temperature input. Joint adds the fault-coupling term to Belief. Matched budget is Belief with a running cap on the share of frames that use C1, set to the share THR used for the same seed and ambient temperature. It has the same heavy-model budget as THR and no temperature reading, which lets us check whether THR helps by choosing good moments for C1 or simply by using C1 less.

5. Experimental Setup

Traces. We use seven real traces with the frontiers and fault labels profiled in [8]. Two come from TartanDrive: the full trajectory, which has no camera faults and serves as a negative control, and a 309-frame stretch of rough terrain. Four come from RADIATE (rain, snow, fog, night). The last is WoodScape fisheye video with real soiling masks. We repeat each trace to fill a 1200-frame mission (2 minutes at 10 Hz) so the chip has time to reach thermal steady state.

Experiments. E1 re-scores the original fault-coupled contention schedule of [8] for bursts of consecutive misses. E2 has thermal contention only, at ambient temperatures of 20 to 55 °C in steps of 5. E3 adds the fault-coupled schedule on top of thermal throttling. In E4, THR is given a wrong R_{th} or τ while the plant keeps the true values.

Protocol. We follow the validation-locked protocol of [9]. Every tunable (plant parameters, H, m, dwell, the 1.5× slow threshold) was set in code before the first evaluation run, and none was changed after seeing results. Each result file records `eval_labels_used_for_selection: false`. All comparisons use five seeds, which change the latency draws, the camera measurement noise and the temperature-sensor noise on the same real frames. We report 95% t-intervals on paired per-seed differences and mark with * any interval that excludes zero.

6. Results

6.1. Cool air

From 20 to 35 °C no policy ever brings the chip to T_{hi}. THR, Belief, Reactive and Matched budget make the same choice on every frame, and their miss rates differ by exactly 0.0 percentage points on all seven traces (Fig. 1). THR only acts differently when its forecast says the chip will cross the threshold. At 40 °C it starts holding C1 back a little early: C1 serves about 93% of frames and pseudo-GT agreement stays between 0.97 and 1.00.

6.2. Hot air

Table 2 lists miss rate, throttling, blackout length, power, energy and accuracy for each policy. Static C1 does badly. At 45 °C it is throttled 20% of the time and has a 12.5 s blackout on every trace. At 55 °C it is throttled 76% of the time, its longest blackout is 91 s, and it misses 88% of deadlines.

Belief and Reactive keep the longest blackout near 1 s, but they are still throttled for 11 to 13% of the mission at 45 °C and 32 to 39% at 55 °C. Fig. 2 shows why. They respond to the slowdown after the governor has already stepped in.

THR was never throttled on any trace, seed or temperature. Compared with Belief it lowers the miss rate by 3.9 to 5.5 percentage points at 45 °C and by 6.5 to 7.4 at 55 °C, and every one of those fourteen differences is significant (Table 3). Energy per on-time frame falls by 11% at 45 °C (3.61 against 4.08 J) and by 13% at 55 °C (2.30 against 2.65 J). THR pays for this with accuracy, since it serves C1 less often: mean pseudo-GT agreement at 45 °C goes from 0.975 to 0.949. On-time utility, which scores a frame by its accuracy if it is on time and zero otherwise, still rises significantly on rain, snow, fog and night at 45 °C (by 2.1 to 4.4 points) and does not change on the other three traces. Static C4 beats every policy on timing and energy, with agreement of only 0.695. THR sits between the two static policies on the accuracy-timeliness trade-off.

TABLE 2. EFFICIENCY, QUALITY AND COST IN E2, AVERAGED OVER 7 TRACES. MISS AND THR. (THROTTLED) IN %. BLACKOUT IS THE LONGEST RUN OF CONSECUTIVE MISSES IN SECONDS. P IS MEAN SOC POWER (W) AND E_{ot} IS ENERGY PER ON-TIME FRAME (J). ACC. IS PSEUDO-GT AGREEMENT (INTERNAL CONSISTENCY ONLY).

Policy	Miss	Thr.	Blackout	P	E_{ot}	Acc.
Ambient 45 °C						
Static C1	60.4	20.3	12.5	23.0	5.80	1.000
Static C4	0.0	0.0	0.0	5.6	0.56	0.695
Reactive	44.4	12.4	1.0	22.6	4.07	0.975
Belief	44.5	12.2	1.0	22.6	4.08	0.975
Matched budget	39.6	0.0	0.9	21.9	3.63	0.948
THR (ours)	39.7	0.0	0.9	21.7	3.61	0.949
Ambient 55 °C						
Static C1	87.7	75.5	90.7	17.4	14.22	1.000
Static C4	0.0	0.0	0.0	5.6	0.56	0.695
Reactive	32.1	37.3	0.9	17.9	2.63	0.927
Belief	32.4	36.3	0.9	17.9	2.65	0.930
Matched budget	25.1	3.7	0.7	17.4	2.32	0.878
THR (ours)	25.4	0.0	0.8	17.2	2.30	0.884

6.3. Matched budget

The matched-budget control uses C1 for the same share of frames as THR, but it picks those frames without knowing the temperature. Its miss rate is statistically indistinguishable from THR's on every trace at both temperatures. The gap is never more than 1.0 percentage point and no interval excludes zero (Table 3). So most of THR's deadline gain comes from how much C1 it runs. Forecasting the right moments adds little on top. Part of the reason is the deadline rule. D is C1's median latency, so C1 misses about half its frames even on a cool chip, and any policy that uses C1 less will miss less.

The two policies do differ on throttling. At 55 °C the matched budget is still throttled on rain (4.5% of the mission), fog (5.8%) and soiling (15.6%), and those paired differences are significant. THR is never throttled. The control also needed THR to tell it the budget in the first place. The right C1 share falls from about 0.93 at 40 °C to between 0.48 and 0.56 at 55 °C, so a fixed cap tuned at one temperature would waste accuracy or throttle at another. For an efficiency-minded deployment, the forecast in THR works as a cheap way to estimate the heavy-model budget online. A duty cap that lands on the same budget by other means should do equally well on misses.

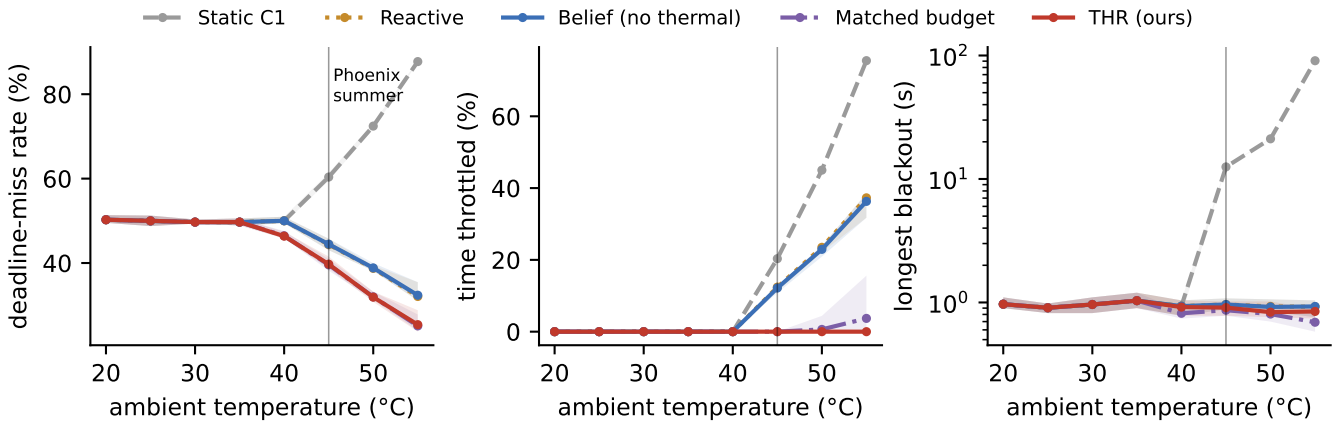


Figure 1. Ambient sweep (E2). Lines are the mean over seven traces and bands span the lowest and highest trace. Left: deadline-miss rate. Middle: share of the mission spent throttled. Right: longest blackout in seconds, log scale. THR (red) never throttles. Matched budget (purple) has the same miss rate as THR but still throttles at 50 and 55 °C.

TABLE 3. HEAT REGIME (E2) PER TRACE. BLACKOUT C1 IS STATIC C1'S LONGEST RUN OF CONSECUTIVE MISSES (S). THR. IS THE % OF THE MISSION SPENT THROTTLED. Δ IS THE REDUCTION IN DEADLINE-MISS RATE RELATIVE TO BELIEF, IN PERCENTAGE POINTS (POSITIVE MEANS FEWER MISSES). THR−MB COMPARES THR WITH THE MATCHED-BUDGET CONTROL. *: THE 95% CI OVER 5 SEEDS EXCLUDES ZERO.

	Ambient 45 °C							Ambient 55 °C					
Trace	Blackout C1	Thr. Belief	Thr. THR	Δ THR	Δ MB	THR−MB		Blackout C1	Thr. Belief	Thr. THR	Δ THR	Δ MB	THR−MB
Off-road (clean)	12.5	13	0	+4.8*	+4.3*	+0.5	90.7	38	0	+6.8*	+7.2*	-0.4	
Off-road seg.	12.5	12	0	+4.6*	+5.2*	-0.6	90.7	37	0	+6.5*	+6.6*	-0.1	
Rain	12.5	13	0	+5.1*	+5.4*	-0.3	90.8	36	0	+7.3*	+7.8*	-0.5	
Snow	12.6	12	0	+5.5*	+5.9*	-0.3	90.7	38	0	+6.8*	+7.3*	-0.5	
Fog	12.4	11	0	+4.8*	+5.0*	-0.2	90.7	34	0	+7.4*	+7.2*	+0.2	
Night	12.5	13	0	+4.3*	+3.9*	+0.4	90.7	38	0	+7.3*	+7.1*	+0.2	
Soiling (fisheye)	12.5	11	0	+3.9*	+4.5*	-0.6	90.7	32	0	+6.7*	+7.7*	-1.0	

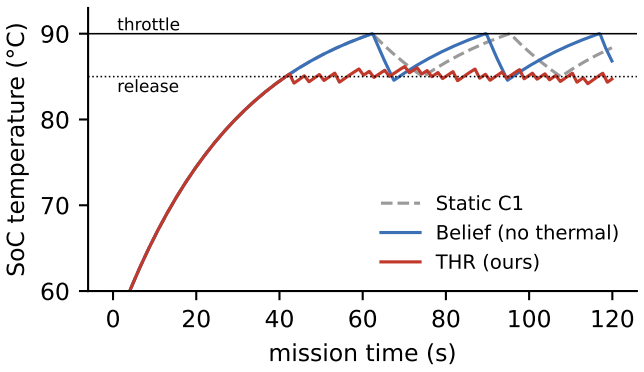


Figure 2. SoC temperature during one 2-minute mission at 45 °C (fog trace, seed 0). Static C1 and Belief go through repeated throttle cycles. THR alternates C1 and C2 and stays just below the release threshold.

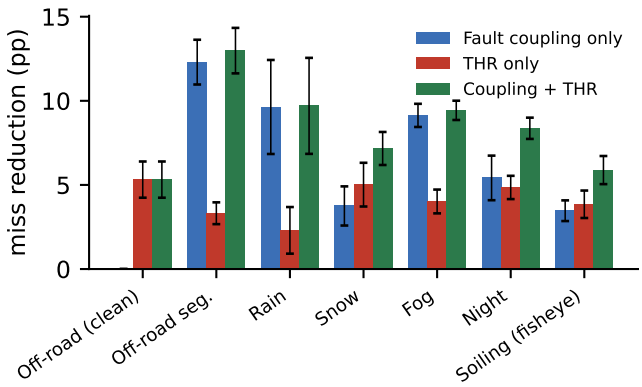


Figure 3. Heat plus fault-coupled contention at 45 °C (E3). Bars show the reduction in deadline-miss rate relative to Belief, with 95% CIs. Joint + THR is best or tied on every trace.

6.4. Heat combined with camera faults

E3 adds fault-coupled contention on top of throttling (Fig. 3, Table 4). The fault-coupling term and THR respond to different problems. On the fault-free trajectory the coupling term (Joint) gains exactly 0.0 points, and THR gains 5.3 points at 45 °C and 8.6 at 55 °C. At 25 °C THR adds nothing, and Joint + THR matches Joint exactly. At 45 and 55 °C, Joint + THR beats Belief on all seven traces, by 5.3 to 13.0 points and by 6.9 to 13.7 points, and it never falls below Joint alone. At 55 °C Joint by itself is no longer significant on snow and soiling, while Joint + THR is (9.0 and 6.9 points).

TABLE 4. E3: REDUCTION IN MISS RATE RELATIVE TO BELIEF (PP). J IS JOINT (FAULT COUPLING ONLY); J+T IS JOINT + THR.

	25 °C		45 °C		55 °C	
Trace	J	J+T	J	J+T	J	J+T
Off-road (clean)	+0.0	+0.0	+0.0	+5.3*	+0.0	+8.6*
Off-road seg.	+12.6*	+12.6*	+12.3*	+13.0*	+9.5*	+13.7*
Rain	+9.9*	+9.9*	+9.6*	+9.7*	+6.2*	+9.9*
Snow	+5.0*	+5.0*	+3.8*	+7.2*	+1.5	+9.0*
Fog	+11.9*	+11.9*	+9.1*	+9.4*	+3.7*	+9.5*
Night	+8.1*	+8.1*	+5.4*	+8.4*	+3.0*	+10.1*
Soiling (fisheye)	+4.7*	+4.7*	+3.5*	+5.9*	+1.0	+6.9*

TABLE 5. THR WITH A WRONG THERMAL MODEL AT 45 °C (E4): RANGE OF THE MISS-RATE REDUCTION RELATIVE TO BELIEF OVER 7 TRACES, NUMBER OF TRACES WITH A SIGNIFICANT REDUCTION, ACCURACY RANGE, AND THE LARGEST THROTTLED SHARE.

THR's model	Δ miss (pp)	Sig.	Acc.	Max thr. (%)
true model	+3.9 to +5.5	7/7	0.921 to 0.994	0
$R_{\mathrm{th}} \times 0.7$	+0.6 to +2.3	5/7	0.941 to 0.996	0
$R_{\mathrm{th}} \times 1.3$	+7.3 to +9.6	7/7	0.890 to 0.994	0
$\tau \times 0.5$	+6.0 to +8.1	7/7	0.899 to 0.994	0
$\tau \times 2$	+3.2 to +4.8	7/7	0.921 to 0.995	0

6.5. Wrong thermal model

A real deployment would have to estimate R_{th} and τ. In E4 (Table 5) we change the model THR believes in while leaving the plant alone, at 45 °C. Because THR forecasts from the measured temperature, no error we tried brought throttling back on any trace. The errors change how cautious THR is. If it thinks R_{th} is 30% lower than it is, the miss-rate gain shrinks to 0.6 to 2.3 points (significant on 5 of 7 traces) and more accuracy is kept. Overestimating R_{th} or underestimating τ gives larger miss reductions and lower accuracy. A rough model is enough to avoid throttling, and the value of R_{th} mostly decides where THR lands between accuracy and timeliness.

6.6. Bursts of misses under fault-coupled contention

A field robot cares about how long it goes without a usable detection, and the average miss rate hides that. We re-scored the original fault-coupled schedule of [8] (E1, Table 6). Fault coupling lowers the share of frames that fall inside bursts of three or more misses (0.3 s) on 4 of 7 traces: by 2.7 points on the off-road stretch, 6.3 on fog, 3.3 on night and 1.0 on soiling. It is exactly zero on the clean trajectory. On no trace does it significantly shorten the longest blackout. Coupling makes bursts less frequent, but the worst case stays about where it was.

7. Discussion and Limitations

In a closed-loop simulation built on real camera traces and real profiled latencies, a closed-form thermal forecast kept the chip out of throttling at a cost of microseconds per frame. It lowered deadline misses and energy per on-time frame in hot air, changed nothing below 40 °C, and worked alongside fault-aware routing. The matched-budget result limits what we can claim. A duty cap with no temperature input does as well on misses once it has the same budget, so the finding is about budgeting heavy inference. We have not shown that THR's choice of when to run C1 matters on its own.

TABLE 6. BURSTS UNDER FAULT-COUPLED CONTENTION (E1). BURST SHARE IS THE % OF FRAMES INSIDE RUNS OF AT LEAST 3 MISSES. MAX RUN IS IN FRAMES.

	Burst share (%)			Max run	
Trace	Belief	Joint	Δ (pp)	Belief	Joint
Off-road (clean)	20.1	20.1	+0.0	7.2	7.2
Off-road seg.	16.5	13.8	+2.7*	8.0	6.4
Rain	12.9	12.3	+0.6	5.6	5.6
Snow	20.8	18.9	+1.9	6.8	7.2
Fog	22.3	16.1	+6.3*	6.8	7.0
Night	18.8	15.5	+3.3*	8.2	8.0
Soiling (fisheye)	14.2	13.1	+1.0*	10.8	10.6

The simulation has clear gaps. The plant is one RC node with typical parameters that we did not measure. Throttling reuses the profiled contended latency distribution instead of real DVFS states. Latencies were measured on an Apple M5 GPU, not an embedded module. Traces are repeated to reach mission length. Accuracy is agreement with the largest model, which tracks real annotations poorly [8], so accuracy and utility numbers only show internal consistency. Seeds change noise on fixed frames and do not stand in for independent missions.

Next, we plan to measure R_{th}, τ and throttle behaviour on a Jetson Orin inside an environmental chamber from 25 to 55 °C using the `tegrastats` harness, and to score the routers against real annotations. We would also like to learn the duty budget directly from temperature logs, which would remove the need for an RC model.

8. Conclusion

On a hot robot, choosing a heavy model now can cause throttling a minute later. THR gives the router a closed-form temperature forecast that costs microseconds per frame. Across seven real camera traces at 45 and 55 °C the chip never throttled under THR. Deadline misses dropped by 3.9 to 7.4 percentage points and energy per on-time frame by 11 to 13%, while cool-weather behaviour stayed the same. The matched-budget control shows that the share of heavy-model frames is what drives the miss-rate gain. Teams building perception for hot hardware should estimate that share online from temperature instead of fixing it ahead of time.

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